

PYTHON DATA ANALYSIS PROJECT

Global Life Expectancy & Healthy Life Expectancy

Exploratory Data Analysis of Countries, Genders and WHO Regions, 2000–2021

INDUSTRY

Healthcare & Public Health

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Industry Overview

Selected Industry

Healthcare & Public Health

Focus area: population health outcomes measured by life expectancy at birth and healthy life expectancy (HALE).

Brief Introduction

- Health systems, governments and agencies like WHO work to help people live longer and healthier lives
- Life expectancy counts expected years of life at birth
- Healthy life expectancy counts the years expected in good health
- Both are tracked per country, sex and year for comparison

Why Data Analysis Matters

- Reveals long-term progress and setbacks
- Exposes inequality between countries, sexes and regions
- Separates extra years lived from years lived in good health
- Guides where health resources should be prioritised

Problem Statement

The Problem

- Life expectancy and healthy life expectancy vary considerably across countries, genders and WHO regions
- A longer life does not always mean more healthy years: a gap exists between the two measures

Why It Matters

- The gap is the number of years expected to be lived in less than full health
- Large differences between countries show unequal health outcomes
- Global trends can reverse: the 2020–21 dip shows how fragile gains are

Who Is Affected

- Populations in countries with the lowest life expectancy
- Men and women, whose outcomes differ across countries
- Policymakers, health ministries and global health agencies who plan and fund care

Project Objectives

1

Track trends

Analyse how life expectancy and healthy life expectancy changed from 2000 to 2021

2

Rank countries

Identify countries with the highest and lowest values in the latest year (2021)

3

Compare genders

Measure the male–female difference in both indicators

4

Quantify the health gap

Create and analyse the gap between life expectancy and healthy life expectancy

5

Compare regions

Compare WHO regions and test the relationship between the two indicators

Proposed Solution & Analysis Questions

Proposed Solution

Merge WHO life expectancy and healthy life expectancy data into one clean dataset, then explore it with Python to expose patterns across time, countries, sexes and regions.

How Python Helps

- Pandas: merge, clean, group and pivot
- NumPy: numeric support for calculations
- Matplotlib: charts that make patterns visible
- Jupyter Notebook: reproducible, documented steps

Q1

How has life expectancy and healthy life expectancy changed from 2000 to 2021?

Q2

Which countries have the highest and lowest life expectancy and healthy life expectancy in 2021?

Q3

How do males and females differ in life expectancy and healthy life expectancy?

Q4

What is the gap between life expectancy and healthy life expectancy, and where is it largest?

Q5

How does life expectancy vary across WHO regions?

Q6

How strongly are life expectancy and healthy life expectancy related?

Dataset Information

Dataset Name	Global Life Expectancy & Healthy Life Expectancy (Cleaned)
Dataset Source	WHO Global Health Observatory: life expectancy (WHOSIS_000001) and healthy life expectancy (WHOSIS_000002) at birth
Rows	12,936
Columns	28 (27 merged + Health_Gap)
File Format	CSV
Coverage	2000–2021 · 196 geographic entities (183 countries) · Male, Female, Total

Important Features

GEO_NAME_SHORT

country / region name

DIM_GEO_CODE_TYPE_life

COUNTRY, WHO region, income group...

DIM_TIME

year (2000–2021)

DIM_SEX

MALE, FEMALE, TOTAL

AMOUNT_N_life

life expectancy (years)

AMOUNT_N_healthy

healthy life expectancy (years)

Health_Gap

life – healthy life expectancy

Each row is one place–year–sex observation, with 95% uncertainty bounds (_NL, _NU) for both indicators.

Data Collection & Understanding

Collection & Source

- Two WHO Global Health Observatory CSVs loaded: life expectancy and healthy life expectancy, each $12,936 \times 15$
- Merged on country, year and sex into one dataset

Initial Understanding

- Data types: year and M49 code are int64; AMOUNT_N / NL / NU are float64; the rest are object
- Missing values: 20 rows with blank uncertainty bounds (Antigua and Barbuda, Iceland, Luxembourg); none in the main measures
- Duplicate records: 0 in either file
- Life expectancy mean 70.17 years (min 36.6, max 87.4)

```
print("Life expectancy raw:", life_raw.shape)
print("Healthy life expectancy raw:", healthy_raw.shape)
```

```
Life expectancy raw: (12936, 15)
Healthy life expectancy raw: (12936, 15)
```

```
print(life.dtypes)
```

```
DIM_TIME          int64
DIM_GEO_CODE_M49  int64
GEO_NAME_SHORT    object
DIM_SEX           object
AMOUNT_N          float64
```

```
print(life.duplicated().sum(),
      healthy.duplicated().sum())
```

```
0 0
```

```
print(missing[missing > 0])
```

```
AMOUNT_NL_life      20
AMOUNT_NU_life      20
AMOUNT_NL_healthy   20
AMOUNT_NU_healthy   20
```

Data Cleaning & Transformation

- Standardised column names (stripped whitespace)
- Prepared merge keys: trimmed country names, upper-cased sex, year as numeric
- Checked duplicates: 0 rows, 0 repeated country–year–sex rows
- Inner merge of both files on country, year, sex: $12,936 \times 27$
- Validated categories: 3 sex values, matching geography types and names
- Handled missing values: 20 blank bounds in 4 columns filled with the column median
- Created column Health_Gap = life – healthy life expectancy (28 columns)
- Filtered by geography type and sex per question; saved cleaned CSV

```
df = pd.merge(life, healthy,  
              on=["GEO_NAME_SHORT", "DIM_TIME", "DIM_SEX"],  
              how="inner",  
              suffixes=("_life", "_healthy"))
```

Merged dataset shape: (12936, 27)

```
df["Health_Gap"] = (df["AMOUNT_N_life"]  
                   - df["AMOUNT_N_healthy"])
```

AMOUNT_N_life	AMOUNT_N_healthy	Health_Gap
74.334559	64.675751	9.658807
76.455919	66.567954	9.887964

```
for col in bound_cols:  
    df[col] = df[col].fillna(  
        df[col].median())
```

Missing values after filling: 0

Final shape: (12936, 28)

Exploratory Data Analysis

66.9 → 71.2

Avg life expectancy, 2000 → 2021
(+4.31 yrs)

58.3 → 61.8

Avg healthy life expectancy (+3.54
yrs)

9.39 yrs

Avg life vs healthy gap, 2021

r = 0.995

Life vs healthy life expectancy

Key Findings

- Libraries: Pandas for groupby / pivot / corr, NumPy as numeric support, Matplotlib for plots
- Both measures rose until 2019, then fell: life expectancy 72.56 (2019) → 71.23 (2021)
- Rwanda gained most since 2000 (+20.6 yrs)
- 2021 life expectancy: Japan 84.5 vs Lesotho 51.5
- 2021 average: women 73.7 vs men 68.8 years (gap 4.84); healthy years gap 2.04

```
life = (df[(df["DIM_GEO_CODE_TYPE_life"]  
          == "COUNTRY") & (df["DIM_SEX"] == "TOTAL")]  
        .groupby("DIM_TIME")["AMOUNT_N_life"]  
        .mean())
```

2000	66.915183
2010	70.248527
2019	72.559107
2020	71.993142
2021	71.226627

```
relationship["AMOUNT_N_life"].corr(  
    relationship["AMOUNT_N_healthy"])
```

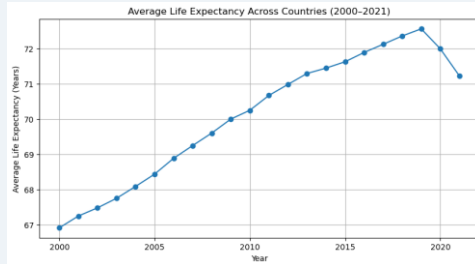
Correlation: 0.9951845506028506

```
health_gap["Health_Gap"].mean()
```

Average health gap: 9.386893050710382

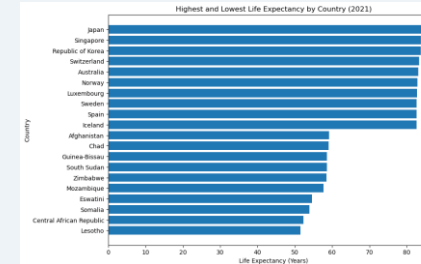
Data Visualization

Line chart: Avg Life Expectancy 2000–2021



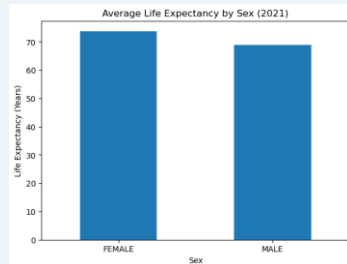
Purpose: Show the trend over time
Shows: Rises from 66.9 to 72.6 (2019), then dips
Insight: 2020–21 wiped out about 1.3 years of gain

Bar chart: Top & Bottom 10 Countries (2021)



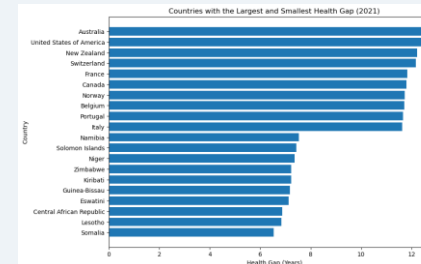
Purpose: Compare extremes in life expectancy
Shows: Japan 84.5 at the top, Lesotho 51.5 at the bottom
Insight: A gap of about 33 years between countries

Bar chart: Avg Life Expectancy by Sex (2021)



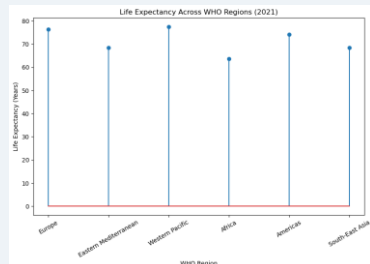
Purpose: Compare males and females
Shows: Women 73.7 years, men 68.8 years
Insight: Women live about 4.8 years longer on average

Bar chart: Largest & Smallest Health Gap (2021)



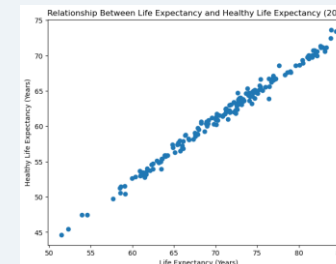
Purpose: Show years lived in less than full health
Shows: Australia 12.5 years vs Somalia 6.5 years
Insight: Long-lived countries have the biggest gaps

Stem plot: Life Expectancy by WHO Region (2021)



Purpose: Compare WHO regions
Shows: Western Pacific 77.4 highest, Africa 63.6 lowest
Insight: Africa trails the top region by about 13.9 years

Scatter plot: Life vs Healthy Life Expectancy (2021)



Purpose: Test the relationship
Shows: Points form a near-straight upward line ($r = 0.995$)
Insight: Healthy years rise almost in step with life expectancy

Key Insights & Recommendations

Key Insights

- Life expectancy rose 4.31 years (66.9 → 71.2) and healthy life expectancy 3.54 years (58.3 → 61.8) from 2000 to 2021
- Both peaked in 2019 and fell in 2020–21: life expectancy lost about 1.3 years
- Biggest gains: Rwanda +20.6, Burundi +19.6, Malawi +17.8 years
- 2021 extremes: Japan 84.5 vs Lesotho 51.5 (life); Singapore 73.6 vs Lesotho 44.6 (healthy)
- Women outlive men by 4.84 years, but healthy-years advantage is only 2.04
- Average health gap is 9.39 years; largest in Australia and the USA (~12.5), smallest in Somalia (6.5)
- WHO regions: Western Pacific 77.4 highest, Africa 63.6 lowest
- Life and healthy life expectancy are almost perfectly correlated ($r = 0.995$)

Recommendations

- Focus resources on the Africa region and the lowest-ranked countries
- Track healthy life expectancy, not life expectancy alone
- Plan for the growing gap as populations live longer
- Study what drove Rwanda, Burundi and Malawi gains, and review post-2019 recovery

Conclusion

What Was Achieved

- Merged two WHO datasets into a clean $12,936 \times 28$ dataset
- Answered 10 EDA questions with Pandas and Matplotlib

Major Findings

- Health improved from 2000 to 2019, then dipped
- Large country, region and gender differences
- Longer lives come with a wider health gap

Addressing the Problem

- Pinpoints where life and healthy years are lowest
- Shows the gap that policy should target

Future Scope

- Add factors such as health spending and income
- Use Seaborn / dashboards and forecasting models
- Analyse trends by income group and sex over time